

Right-to-Repair & Circular Economy Platform

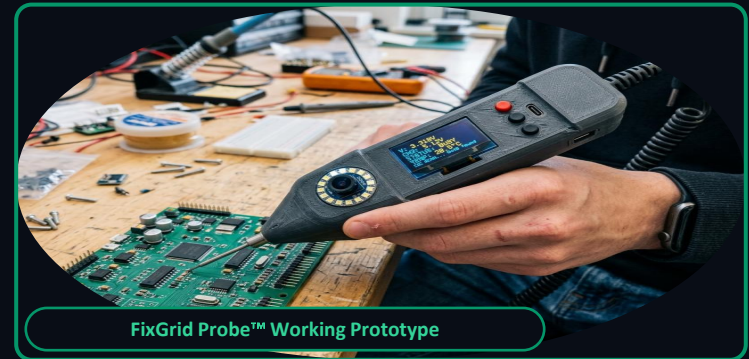
FixGrid Probe™

The Complete Ecosystem: A Live Web Platform (fixgrid.vytron.me) & A Frugal Handheld Diagnostic Wand Championing Sustainable Repair Across All 21+ Categories

✓ Live Digital Platform: Search & escrow across 21+ categories (Appliances, Tech, EVs)

✓ Physical Innovation: Under ₹1,800 FixGrid Probe™ sub-junction diagnostic wand

✓ Dual-Loop QR Warranty: Tamper-evident seals with photo proof



Problem Statement: The Throwaway Culture & Climate Paradox

1. The Trust Void Across All 21+ Categories

- 78% of consumers discard repairable goods (appliances, consumer tech, tools) fearing arbitrary pricing & zero warranty.
- Spans ALL 21+ categories: Major home appliances (ACs, fridges, washing machines), laptops, phones, TVs & EVs.
- Local repair artisans across every trade remain digitally invisible, informal, and overlooked.

2. The Universal Hardware Diagnostic Void

- Lab diagnostic equipment costs ₹50,000–₹1.5L—unaffordable for neighborhood appliance & tech workshops.
- Standard multimeters inject 2.5V–3.0V, risking delicate micro-controllers in modern smart appliances & boards.
- Without visual proof, mechanics guess—declaring entire appliance control boards or devices "unfixable."

₹1.2L Cr

THE LOST VALUE

Consumers waste billions replacing appliances & devices when an artisan 500m away could repair it.

85%+

SKILLED HEROES OVERLOOKED

Informal repair mechanics across 21+ categories lack digital tools and are trapped in unverified guesswork.

Millions

E-WASTE CATASTROPHE

Heavy appliances & electronics flood landfills, leaching toxic metals while factories burn carbon for replacements.

Proposed Solution: The FixGrid Hardware-Software Innovation

PILLAR 1: DIGITAL PLATFORM

FixGrid Trust & Warranty Cloud Engine

- All 21+ Categories: Covers Home Appliances, Consumer Tech, Power Tools, EVs & Wearables.
- Upfront Smart Escrow: Customer payments locked safely at booking; released on verified sign-off.
- 90-Day Digital Warranty: Platform-guaranteed warranty passport verifiable on any smartphone.
- Artisan Empowerment: Digital storefronts, verified reviews & 5% completed-job rebate.

PILLAR 2: HARDWARE WAND

FixGrid Probe™ Multimodal Diagnostic Tool

- 0.40V Sub-Junction Divider: Clamps below 0.6V P-N junction—tests in-circuit without desoldering.
- 16-Bit ADS1115 + Tone: Pitch-shift audio buzzer (1.2kHz → 2.4kHz) guides to shorted 0402 SMDs.
- ESP32-CAM Microscope: 2.5cm macro lens & shadowless ring LEDs captures HD fault photo proof.
- Hardware Trust-Gate: Transmits cryptographic VERIFIED_PASS token to release escrow funds.

₹1,795 BOM

Frugal Prototype Cost

0.40V Clamp

Sub-Junction In-Circuit

Dual-Loop QR

Digital Product Passport

Mission LiFE

E-Waste Mitigation Goal

FixGrid Probe™: Lab Diagnostics in a Handheld Wand



FixGrid Probe™ Working Prototype (₹1,795 / ~\$21 BOM)

MICRO-TOUCH SENSING

Instant Fault Chirp

Technician touches probe needle to circuit trace. Conducts rapid impedance testing in milliseconds, beeping and lighting Green (Normal) or Red (Fault) on OLED screen. Safe sub-3.3V (<5mA) sensing prevents static shock.

MAGNIFIED VISUALS

5MP Macro Photo Proof

Integrated 5MP macro camera with polarized ring LEDs captures 10x close-up photos of burned ICs or cracked solder joints, sending undeniable visual proof straight to the customer's smartphone before soldering.

DIGITAL PASSPORT

Dual-Loop QR Warranty

Tool writes a physical tamper-evident QR void seal placed over device or appliance seam. Customer scans to review diagnostic readings, before/after photos, and activates a guaranteed 90-day digital warranty.

FixGrid Web Platform (fixgrid.vytron.me)

Hyper-Local Search & Map Engine

Allows consumers to discover verified repair shops within walking distance across 21+ categories (Major Home Appliances, ACs, Microwaves, Phones, Laptops, Power Tools, EVs) with live status.

- Next.js 16 + Leaflet Map Engine

Smart Escrow & Payment Protection

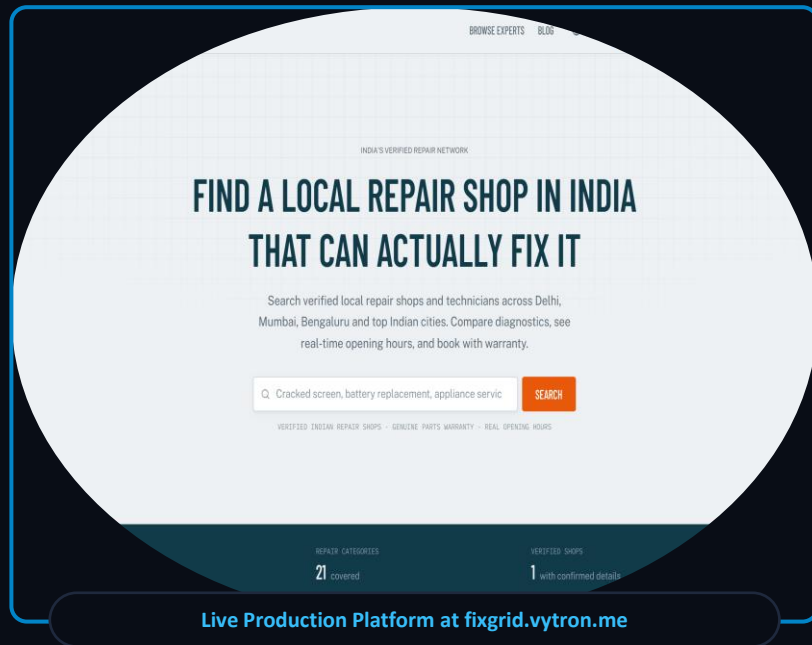
Customer payments are locked securely on the platform upon booking and released to the shopkeeper only after the customer inspects and signs off on verified satisfaction.

- Supabase PostgreSQL Triggers & RLS

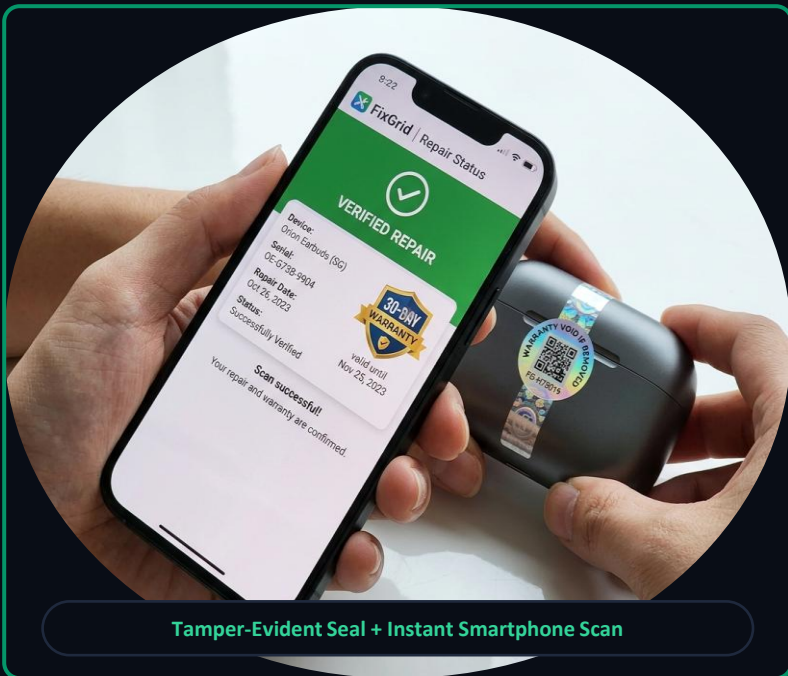
Artisan Storefronts & 5% Cashback

Provides informal repair mechanics with instant SEO-indexed digital profiles, verified reviews, diagnostic log uploads, and a 5% completed-bill cashback incentive.

- Shopkeeper Empowerment Engine



Dual-Loop QR Digital Product Passport



Tamper-Evident Seal + Instant Smartphone Scan

STEP 01

FixGrid Probe Diagnostics Upload

When the technician completes the repair across any category, the FixGrid Probe wand syncs the passing impedance test and high-res macro photos directly to the FixGrid cloud database via ESP32 Wi-Fi/BLE.

STEP 02

Physical Holographic Void Seal

A serialized, tamper-evident holographic QR void sticker is printed and affixed over the device or appliance seam. If anyone attempts to tamper with or reopen the unit, the physical pattern breaks.

STEP 03

Customer Scan & Warranty Activation

The customer scans the QR code with any standard smartphone camera. It displays component before/after photos, diagnostic health logs, and activates an ironclad 90-day digital warranty certificate.

Tapping the \$15.2B Repair Economy

60–80%**ECONOMIC (Consumer & Artisan)**

Consumers save 60–80% compared to purchasing new devices. Local mechanics gain formal digital identity, repeat customers, and higher daily earnings.

100%**SOCIAL (Community Dignity)**

Brings unorganized street artisans into the formal digital economy, elevating informal mechanics with lab-grade diagnostic tools and transparent trust.

Zero**ECOLOGICAL (Planet Circularity)**

Diverts hundreds of kilograms of toxic printed circuit boards and lithium batteries from municipal landfills, cutting manufacturing carbon emissions.

1. Platform Take-Rate (5–8%)

Competitive transaction fee on every escrow-verified repair booking.

2. Customer Care+ Subscription (₹199/mo | ₹1,499/yr)

Household repair pass: free FixGrid Probe diagnostics, zero booking fees & 60-day warranty.

3. Shop Pro SaaS & Hardware Tier (₹999/mo)

FixGrid Probe firmware updates, warranty management, digital store CRM & analytics.

4. Certified Spare Parts Logistics

Curated B2B marketplace revenue for verified authentic micro-components.

A Movement Towards a Sustainable Repair Economy

01

PHASE 1 (MONTHS 1–6)

Pilot Repair Clusters

- Deploy 50 FixGrid Probe units across Mumbai electronics repair markets.
- Validate web escrow & before/after macro photo flow with 1,000+ customer repairs.

02

PHASE 2 (MONTHS 7–12)

Community Drives

- Partner with schools & colleges for zero-e-waste community repair clinics.
- Roll out Customer Care+ subscription pass and certified technician onboarding.

03

PHASE 3 (YEAR 2+)

Municipal Scaling

- Integrate with municipal e-waste boards for certified gadget triage.
- Scale frugal FixGrid Probe tool manufacturing to local repairers nationwide.

FixGrid Probe™ Unified Ecosystem

A Unified Web & Hardware Solution for Right-to-Repair, Consumer Trust & Circular Sustainability

Project FixGrid

fixgrid.vytron.me